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Biosensing test strip and biosensing test method

Abstract

Disclosed are a biosensing test strip (100, 200, 300, 500, 600, 700, 800, 900, 1000, 1100) and a biosensing test method. The biosensing test strip (100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100) comprises: a reaction layer (120, 220, 720, 820) provided with a reaction flow channel (121, 221, 821, 920, 1020); a partition plate layer (130, 230) located above the reaction layer (120, 220, 720, 820) and covering the reaction flow channel (121, 221, 821, 920, 1020); an exhaust layer (140, 240, 540, 640) located above the partition plate layer (130, 230), with the exhaust layer (140, 240, 540, 640) being provided with an exhaust flow channel (141, 241, 550, 650); and a communication hole passing through the partition plate layer (130, 230) to enable the exhaust flow channel (141, 241, 550, 650) to be in communication with the reaction flow channel (121, 221, 821, 920, 1020).

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a biosensor strip, and more particularly to a biosensor strip having a controlled flow for a liquid sample.

DESCRIPTION OF THE PRIOR ART

(2) In vitro medical measurement plays an extremely critical role in the medical industry. By qualitatively and quantitatively measuring the changes in a biological fluid, index information for quick diagnosis and treatment of diseases can be provided. In medical or biochemical testing, the use of test strips has become a common technique.

(3) The prior art discloses a test strip that uses capillary force to absorb a liquid sample into the reaction zone. Because the capillary force of such type of test strips is affected by physical

conditions such as shear stress, cohesion and viscosity of liquid samples, the flow and force of the liquid sample cannot be controlled. Thus, such a type of test strip is only suitable for simple reactions of a single agent, or multiple agents that can only act on a same position. For example, for a design requirement for a variety of agents to react in a long flow channel, such type of test strips are susceptible to non-uniform mixing due to the inability to control the flow.

(4) The prior art also discloses a measurement carrier, which has one side of a channel thereof configured as a sampling port and the other end connected to an external power device (e.g., a pressure pump), and the effect of controlling the flow of a liquid sample is achieved by the external power device. However, the foregoing external pressuring means suffers from drawbacks of being likely to damage samples (e.g., causing blood cell rupture) and requiring high-precision bonding of seams between a flow channel and a power device.

SUMMARY OF THE INVENTION

(5) In view of the issues of the prior art, the present invention provides a biosensor strip with a function for reducing flow of a liquid sample; more specifically, the biosensor strip has an exhaust flow channel that is configured to reduce the flow of a liquid sample provided by a capillary action.

(6) A biosensor strip and a method thereof are provided according to an embodiment of the present invention. The biosensor strip includes a reaction layer having a reaction flow channel; a separation layer located above the reaction layer and covering the reaction flow channel; an exhaust layer located above the separation layer, the exhaust layer having an exhaust flow channel; and a through hole passing through the separation layer to communicate with the exhaust flow channel and the reaction channel.

(7) A method for determining a structure of a biosensor strip is provided according to another aspect of the present invention. The biosensor strip has an exhaust flow channel to reduce the flow of a liquid sample in a reaction flow channel. The method obtains a structural relation between the exhaust flow channel and the reaction flow channel according to resistance parameters of the liquid sample and air.

(8) The present invention further includes other aspects, and these features and advantages of the present invention can be better understood with reference to the accompanying drawings or using the implementations of the present invention in the description below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a biosensor strip according to a first embodiment of the present invention;

(2) FIG. 2 is a schematic diagram of a biosensor strip according to a second embodiment of the present invention;

(3) FIG. 3 is a schematic diagram of a biosensor strip according to a third embodiment of the present invention;

(4) FIG. 4A and FIG. 4B are schematic diagrams of a choke valve of a biosensor strip according to a fourth embodiment of the present invention;

(5) FIG. 5 is a schematic diagram of a biosensor strip according to a fifth embodiment of the present invention;

(6) FIG. 6A and FIG. 6B are schematic diagrams of a biosensor strip according to a sixth embodiment of the present invention;

(7) FIG. 7 is a schematic diagram of a biosensor strip according to a seventh embodiment of the present invention;

(8) FIG. 8 is a schematic diagram of a biosensor strip according to an eighth embodiment of the present invention;

- (9) FIG. 9A, FIG. 9B and FIG. 9C are schematic diagrams of operating a conventional biosensor strip without an exhaust flow channel;
- (10) FIG. 10A, FIG. 10B, FIG. 100, FIG. 10D, FIG. 10E and FIG. 10F are schematic diagrams of operating a biosensor strip according to an embodiment of the present invention; and
- (11) FIG. 11A, FIG. 11B and FIG. 11C are schematic diagrams of operating a biosensor strip according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(12) Preferred embodiments of the present invention are given as examples with reference to the accompanying drawings below. To avoid obscuring the contents of the present invention, details of conventional elements, associated materials and associated processing techniques are omitted from the description below.

First Embodiment

(13) Referring to FIG. 1, a method for determining a structure of a biosensor strip is provided according to a first embodiment of the present invention. The method is suitable for a biosensor strip **100** in FIG. 1 showing a lateral section diagram. As shown, the biosensor strip **100** is strip-shaped, and is defined with a sampling end **101**. The biosensor strip **100** includes a reaction layer **120** above a substrate layer **110**, the reaction layer **120** having a reaction flow channel **121**; a separation layer **130**, located above the reaction layer **120** and covering the reaction flow channel **121**, the reaction flow channel **121** being a space for a liquid sample **190** to flow and perform an electrochemical reaction; an exhaust layer **140**, located above the separation layer **130**, the exhaust layer **140** having an exhaust flow channel **141**; and a through hole **150**, passing through the separation layer **130** to communicate with the exhaust flow channel **141** and the reaction flow channel **121**. The biosensor strip **100** further includes an upper cover **160** having an exhaust hole **161** for discharging into outside, the exhaust hole **161** being in communication with the exhaust flow channel **141**. The reaction flow channel **121** has a capillary structure. The exhaust flow channel **141** has a function of reducing the flow of the liquid sample **190** provided by a capillary action. It should be noted that, this embodiment provides an illustrative example using an electrochemical strip. In other embodiments, an optical detection window can be provided on the strip. In yet other embodiments, both optics and electrochemistry can be used on the strip.

(14) The method for determining a structure of a biosensor strip according to the first embodiment includes (1) providing the foregoing biosensor strip **100**; (2) calculating a dimensionless resistance ratio μ^* of the biosensor strip **100** according to formula I:

$$(15) \quad \mu^* = \frac{G}{L - G} \quad \text{formula I}$$

where $\mu \cdot \text{sub} \cdot G$ is the resistance of air in the exhaust flow channel, and $\mu \cdot \text{sub} \cdot L$ is the resistance of a liquid sample **190** used in the biosensor strip; (3) determining a numeral value greater than the dimensionless resistance ratio μ^* ; and (4) considering the numeral value as a dimensionless length ratio L^* , and substituting L^* into formula II below to obtain a structural relation between the reaction flow channel **121** and the exhaust flow channel **141**:

$$(16) \quad L^* = \frac{L_L w_G h_G^3 C_G}{L_G w_L h_L^3 C_L}, \quad \text{formula II}$$

where $w \cdot \text{sub} \cdot L$ is wall width of the reaction flow channel **121**, $h \cdot \text{sub} \cdot L$ is the wall height of the reaction flow channel **121**, $L \cdot \text{sub} \cdot L$ is the length of the reaction flow channel **121**, $C \cdot \text{sub} \cdot L$ is a rectangle correction coefficient of the reaction flow channel **121**, $w \cdot \text{sub} \cdot G$ is the wall width of the exhaust flow channel **141**, $h \cdot \text{sub} \cdot G$ is the wall height of the exhaust flow channel **141**, $L \cdot \text{sub} \cdot G$ is the length of the exhaust flow channel **141**, and $C \cdot \text{sub} \cdot G$ is a rectangle correction coefficient of the exhaust flow channel **141**; wherein, $C \cdot \text{sub} \cdot L$ or $C \cdot \text{sub} \cdot G$ are respectively obtained according to formula III below:

$$(17) \quad C = [1 - \frac{1}{n} \cdot \text{Math} \cdot \frac{192}{5} \frac{h}{w} \frac{1}{n^5} \tanh(n \frac{w}{2h})], \quad \text{formula III}$$

where w is the wall width of the reaction flow channel **121** or the exhaust flow channel **141**, and h is the wall height of the reaction flow channel **121** or the exhaust flow channel **141**. Sources of the formulae above are explained below. It should be noted that the method above takes the biosensor strip **100** as an example. The method above is also suitable for other biosensor strips according to other embodiments of the present invention.

(18) The volume of the liquid sample **190** that the biosensor strip **100** can accommodate depends on the shape of the reaction flow channel **121**. The reaction flow channel **121** is an elongated pipeline surrounded on four sides, and an approximate formula the flow $Q_{sub.L}$ of the liquid sample **190** therein is:

$$(19) Q_L = \frac{P w_L h_L^3}{12 \mu_L L_L},$$

where w_L is the wall width of the reaction flow channel **121**, $h_{sub.L}$ is the wall height of the reaction flow channel **121**, μ_L is the viscosity of the liquid sample **190**, $L_{sub.L}$ is the length of the reaction flow channel **121**, and ΔP is the capillary pressure of the liquid sample **190** in the reaction flow channel **121**. Without considering the pressure difference of the distance of the flow channel, the calculation formula for $\Delta P_{sub.L}$ is approximately:

$$(20) P_L = \frac{\cos \theta (2w_L + 2h_L)}{w_L h_L},$$

where γ is the liquid surface tension, and θ is the contact angle of the liquid sample **190** with respect to the surface of the reaction flow channel **121**.

(21) The flowing of the air in the biosensor strip **100** is mainly caused by the force exerted by the liquid sample **190**. In other words, while capillary action is applied, the flowing of the liquid sample **190** in the reaction flow channel **121** needs to push air in order to move forward. An arrow fin FIG. 1 shows the flowing direction of pushed air. Under the effects applied by the wall of the reaction flow channel **121** onto the liquid sample **190** and air, given that the reaction flow channel **121** connects the exhaust flow channel **141** in series and that the air flow $Q_{sub.G}$ is significantly smaller than the flow $Q_{sub.L}$ of the liquid sample **190**, the air flow controls the flow of the liquid sample **190**. In brief, the flowing speed of air can be controlled by air resistance generated by the exhaust flow channel **141** while the flow of the liquid as well as the bio(electro)chemical reaction time can be controlled simultaneously.

(22) A calculation formula of the air flow in the inner wall of a common exhaust flow channel is:

$$(23) Q_G = \frac{P w_G h_G^3 C}{12 \mu_G L_G},$$

(24) where $w_{sub.G}$ is the wall width of the exhaust flow channel, $h_{sub.G}$ is the wall height of the exhaust flow channel, $\mu_{sub.G}$ is the viscosity of air, $L_{sub.G}$ is the length of the exhaust flow channel, and C is a rectangle correction coefficient, which is the resistance coefficient generated by a rectangle flow channel with respect to flowing of air and can be calculated by the formula below:

$$(25) C = [1 - \sum_{n, \text{ odd}}^{\infty} \frac{192}{n^5} \frac{h}{w} \frac{1}{n^5} \tanh(n \frac{w}{2h})].$$

(26) The exhaust flow channel has a lateral cross section perpendicular to an extension direction of the exhaust flow channel. To calculate by bringing various length-to-width ratios of different lateral cross sections into the above formula of the rectangle correction coefficient, it is obtained that an exhaust flow channel with rectangular cross section has less influence than that with a square cross section. Thus, it is deduced that the exhaust flow channel preferably has a lateral cross section having a long diameter and a short diameter. In a preferred embodiment, a ratio of the long diameter to the short diameter is greater than 2, more preferably greater than 6, and most preferably greater than 8.

(27) It is known from the calculation formula of air flow above that, the length of the exhaust flow channel **141** is crucial to producing effect of air resistance upon the liquid sample **190**. As shown in FIG. 1, the volume of the exhaust flow channel **141** is V_2 , and the volume of the reaction flow channel **121** is V_1 . Before the liquid sample **190** is injected to the sampling port **101**, air fills the reaction flow channel **121** and the exhaust flow channel **141**. The liquid sample **190** is affected by

the air resistance from the moment it enters the reaction flow channel **121** caused by the resulted capillary action. Thus, the volume V1 of the reaction flow channel **121** and the volume V2 of the exhaust flow channel **141** are taken into account when the overall effect of air resistance is calculated. Therefore, the calculation formula for the total air flow of the biosensor strip **100** should be:

$$(28) Q_{GT} = \frac{Q_{GV1} Q_{GV2}}{Q_{GV1} + Q_{GV2}},$$

where Q.sub.GT is the total air flow, Q.sub.GV1 is the air flow of the reaction flow channel **121**, and Q.sub.GV2 is the air flow of the exhaust flow channel.

(29) However, because the cross-section area of the reaction flow channel **121** is far greater than the cross section of the reaction flow channel **141**, the air flow of the reaction flow channel **121** is also driven by the air flow of the exhaust flow channel **141**, such that the air flow Q.sub.GV1 of the reaction flow channel **121** can be omitted in the calculation for the total air flow Q.sub.GT.

Therefore, the calculation formula for the total air flow can be corrected to:

$$Q_{sub.GT} \approx Q_{sub.GV2}.$$

(30) The viscosities of various liquid samples will change the air flow Q.sub.G. Therefore, in this embodiment, in order to minimize the influence on the air flow Q.sub.G caused by the viscosities of various liquid samples, the relation of the resistance contributed by the flow channels, the viscosity of the liquid sample and the viscosity of air should be considered, as follows:

$$(31) R_G > R_L \cdot \frac{12 \cdot \frac{L_G}{w_G h_G^3 C_G}}{12 \cdot \frac{L_L}{w_L h_L^3 C_L} \cdot \frac{L_G}{L_L}} > \frac{L_L w_G h_G^3 C_G}{L_G w_L h_L^3 C_L} \cdot \mu^* > L^*,$$

where R.sub.G is the resistance of air in the exhaust flow channel **141**, ΔR.sub.L is the resistance difference between the liquid sample **190** and the air in the reaction flow channel **121**, C.sub.G and C.sub.L are respectively rectangle correction coefficients of the exhaust flow channel **141** and the reaction flow channel **121**, μ* is a dimensionless resistance ratio, and L* is a dimensionless length ratio.

(32) It is known from the above, the biosensor strip **100** satisfying the condition of the dimensionless resistance ratio μ* being greater than the dimensionless length ratio L* ensures that the resistance generated by the air and the liquid sample **190** in the reaction flow channel **121** is smaller than the resistance generated by air in the exhaust flow channel **141**, thereby achieving the air flow Q.sub.G far smaller than the liquid sample flow Q.sub.L. However, the present invention does not define the difference between μ* and L*, preferably, the ratio difference between μ* and L* is 5 times, more preferably, the ratio difference between μ* and L* is 20 times, and most preferably, the ratio difference between μ* and L* is 100 times. In another embodiment, to reduce the interference of ambient temperature on the control for the flow of the liquid sample **190**, a heater applied for the biosensor strip **100** can be configured in a measuring device (not shown), so as to ensure that each time the biosensor strip **100** is used, the pressure in the reaction flow channel **121** or the exhaust flow channel **141** stays the same.

(33) The formulae above are suitable for some embodiments. However, for some other embodiments, for example, when the viscosity of a liquid sample is extremely large such that the formulae above are inapplicable, adjustment can be made by, for example, the method of the second embodiment below.

Second Embodiment

(34) FIG. 2 shows an exploded schematic diagram of the components of a biosensor strip **200** according to the second embodiment. The biosensor strip **200** is strip-shaped, and is defined with a sampling end **201** and a connecting end **202**, wherein the connecting end **202** is for electrically connecting to a measuring device (not shown). Referring to FIG. 2, the biosensor strip **200** includes a reaction layer **220** above a substrate layer **210**, the reaction layer **220** having a reaction channel **221**; a separation layer **230**, located above the reaction layer **220** and covering the reaction flow channel **221**; an exhaust layer **240**, located above the separation layer **230**, the exhaust layer **240** having an exhaust flow channel **241**; and a through hole **250** passing through the separation layer

230 to communicate with the exhaust flow channel **241** and the reaction flow channel **221**. The biosensor strip **200** further includes an upper cover **260** having an exhaust hole **261** for discharging into outside and communicating with the exhaust flow channel **241**. In another embodiment, the exhaust hole (not shown) is provided on the exhaust layer **240**. The design details (for example, sizes of the channel or holes for the components) with respect to the reaction flow channel **221**, the through hole **250** and the exhaust flow channel **241** can be referred from the method of the first embodiment. In this embodiment, the reaction flow channel **221** is also a capillary structure, and the exhaust flow channel **241** has a function of reducing the flow of the liquid sample **190** provided by a capillary action.

(35) Referring to FIG. 2, the substrate layer **210**, the reaction layer **220**, the separation layer **230**, the exhaust layer **240** and the upper cover **260** above are formed in general of an insulation material, and can be one selected from a group consisting of polyvinyl chloride (PVC), glass fiber (FR-4), polyester (polyester suphone), bakelite, polyethylene terephthalate (PET), polycarbonate (PC), polypropylene (PP), polyethylene (PE), polystyrene (PS), glass plate and ceramic, or any combination thereof. The layers are sequentially combined as shown to form the biosensor strip **200**.

(36) Referring to FIG. 2, the substrate layer **210** further includes an electrode unit **211** configured on the substrate layer **210**. The electrode unit **211** can be consisted of at least one group of mutually insulated electrodes. The material of the electrode unit **211** can be any electrically conductive substance, such as palladium glue, platinum glue, gold glue, titanium glue, carbon glue, silver glue, copper glue, gold and silver mixed glue, carbon silver mixed glue, or any combination of these electrically conductive materials.

(37) Referring to FIG. 2, the reaction flow channel **221** of the reaction layer **220** has a sampling port **222** located on the sampling end **201**. A liquid sample suitable for the present invention is usually blood; however, the present invention is not limited thereto. Four reagent areas **223a**, **223b**, **223c** and **223d** are arranged along the reaction flow channel **221** in the direction of the liquid sample entering the reaction flow channel **221** from the sampling port, and are individually placed with the same or various different reagents **224**. A region having a larger area on the reaction flow channel **221** is for the purpose of moderating the flow of the liquid sample in that region. The liquid sample stays in a region having a larger area for a relatively longer period of time. A reagent can be placed in a region having a larger area; however, the present invention is not limited thereto. The present invention includes other embodiments having various numbers of reagent area, including only one reagent area. The reagent suitable for the present invention includes enzymes (e.g., glucose glucoamylase), conductive media (such as red blood salt), phosphate buffer, protective agents (such as proteins, dextrin, glucosan, amino acids, etc.), and also includes components with biological identification capabilities such as antigens, antibodies, microbial cells, animal and plant cells, and animal and plant tissues. FIG. 2 shows that the reaction flow channel **221** is a linear path; however, the present invention also includes other embodiments in which the reaction flow channel **221** is in other path shapes. As shown in FIG. 2, in this embodiment, when the liquid sample enters the reaction flow channel **221** from the sampling end **222**, air in the reaction flow channel **221** flows toward a terminal end **221t**, and air in the exhaust flow channel **241** also flows toward the direction of the exhaust hole **261**.

(38) Referring to FIG. 2, the separation layer **230** has on the side of the sampling end **201** a first notch **231** corresponding to the sampling port **222**. The through hole **250** passes through the separation layer **230** at a side of near the terminal end **221t**. The separation layer **230**, the inner wall of the reaction layer **220** and the substrate layer **210** jointly form the space of the reaction flow channel **221**, wherein the separation layer **230** constitutes the upper wall of the reaction flow channel **221**, the substrate layer **210** constitutes the bottom wall of the reaction flow channel **221**, and the inner wall of the reaction layer **220** constitutes the sidewall of the reaction flow channel **221**. In one embodiment, at least one wall surface of the reaction flow channel **221**, preferably, at

least one wall surface located on the bottom wall or the upper wall of the reaction flow channel **221**, is hydrophilic. In another embodiment, compared to the bottom wall or the upper wall of the reaction flow channel **221**, the hole wall of the through hole **250** is more hydrophobic. According to FIG. 2, with respect to the starting end (the position of the sampling port **222**) of the reaction flow channel **221**, the through hole **250** corresponds to the terminal end **221t** of the reaction flow channel **221**. That is, air from the reaction flow channel **221** flows through the last reagent area **223d** and then arrives at the through hole **250**. In other embodiments, the position of the through hole **250** can correspond to other appropriate position of the reaction flow channel **221**.

(39) Referring to FIG. 2, the exhaust layer **240** has on the side of the sampling end **201** a second notch **242** corresponding to the first notch **231** and the sampling port **222**. FIG. 2 shows that the exhaust flow channel **241** is distributed in a wave manner on the exhaust layer **240**; however, the present invention also includes embodiments in which the reaction flow channel is in other shapes, including shapes of a straight line, a digital wave, a sawtooth, a triangular wave and various irregular curves.

(40) Referring to FIG. 2, the upper cover **260** has on the side of the sampling end **201** a third notch **262** corresponding to the second notch **242**, the first notch **231** and the sampling port **222**. The upper wall **260**, the inner wall of the exhaust layer **240** and the separation layer **230** jointly form the space of the exhaust flow channel **241**, wherein the upper cover **260** constitutes the upper wall of the exhaust flow channel **241**, the separation layer **230** constitutes the bottom wall of the exhaust flow channel **241**, and the inner wall of the exhaust layer **240** constitutes the sidewall of the exhaust flow channel **241**. In another embodiment, the size of the third notch **262** is smaller than that of the sampling port **222**. In other embodiments, the size of the first notch **231** or the second notch **242** can be smaller than the sampling port **222**.

Third Embodiment

(41) FIG. 3 shows an exploded schematic diagram of the components of a biosensor strip **300** of the third embodiment. The biosensor strip **300** is similar to the biosensor strip **200** of the second embodiment, and differs in that the biosensor strip **300** further includes a hydrophilic layer **310**. The hydrophilic layer **310** has a first exposure opening **311**, a second exposure opening **312** and a third exposure opening **313** for exposing electrodes underneath the substrate layer **210**. The function of the hydrophilic layer **310** is to reduce the contact angle between the liquid sample in the reaction flow channel **221** and the channel surface so as to achieve an effect of stabilizing flow. The third embodiment can be used for cases where the hydrophilic ability of the substrate layer **210** is insufficient in need. For example, the change in hydrophilicity on the surface of the substrate **210** is restrained by placing the necessary electrode unit **211** thereon. In this embodiment, the flow of a common liquid sample in the reaction flow channel **221** is controlled by the exhaust flow channel **241**. However, the flow of certain high-viscosity liquid samples in the reaction flow channel **221** could be slower than the expected flow controlled by the exhaust flow channel, resulting in a flow error in the flow controlled by the exhaust flow channel **241**. Such unfavorable situations can be alleviated or even avoided by means of the hydrophilic layer **310**.

Fourth Embodiment

(42) FIG. 4A and FIG. 4B are section perspective schematic diagrams of a choke valve **410** of a biosensor strip of the fourth embodiment, wherein FIG. 4A does not depict a liquid sample whereas FIG. 4B depicts a liquid sample. Details of the other components of the fourth embodiment can be referred from the second embodiment or the third embodiment. The choke valve **410** is disposed in the through hole **250**. In other words, referring to the second embodiment or the third embodiment, the choke valve **410** is located at the through hole **250** corresponding to the terminal end **221t** of the reaction flow channel **221** and passes through the separation layer **230**. In this embodiment, the choke valve **410** can prevent overflow of the liquid sample from entering the exhaust flow channel **241**.

(43) Referring to FIG. 4A, the choke valve (through hole) **410** has a lower slot **411** in

communication with the reaction flow channel **221**, and a middle slot **412** located above the lower slot **411**. The lower slot **411** has an opening **413** in communication with the middle slot **412**, and the opening **413** is defined with a horizontal plane **413p**. The middle slot **412** has a bottom part **414**, the bottom part **414** has a connecting part **414c** connected to the opening **413**, and a surface of the connecting part **414c** is not higher than the horizontal plane **413p**, such that the liquid sample from the lower slot **411** has no access to a clinging wall surface so as to achieve a flow stop effect. (44) Referring to FIGS. **4A** and **4B**, the size of the middle slot **412** is greater than the size of the lower slot **411**. Moreover, the choke valve **410** further includes an upper slot **415** in communication with the middle slot **412**, and the size of the upper slot **415** is preferably smaller than or equal to that of the middle slot **412**, or preferably smaller than or equal to that of the lower slot **411**. The height of the middle slot **412** is higher than the maximum height of the liquid sample **190** at the opening **413**, so as to further prevent the liquid sample **190** from entering the upper slot **415** or the exhaust flow channel **241**. In other embodiments, the choke valve **410** may exclude the upper slot **412**. In this embodiment, the upper slot **415** of the choke valve **410** is located at the exhaust layer **240**, the middle slot **412** of the choke valve **410** is located at the separation layer **230**, and the lower slot **411** of the choke valve **410** is located at the reaction layer **220**. In another embodiment, the middle slot **412**, the lower slot **411** and the upper slot **415** (if any) can all be located at the separation layer **230**. In another embodiment, the through hole **250** can include two or more vertically overlaid choke valves **410**.

(45) Moreover, in another embodiment, the surface of the connecting part **414c** forms a hydrophobic surface. The flow stop effect of the choke valve **410** can be reinforced by the hydrophobic surface as the surface of the connecting part **414c** is described with reference to FIG. **4B** and the formula below. For example, as the hydrophobic ability of the surface of the connecting part **414c** gets higher, the liquid sample is less likely to approach the surface of the connecting part **414c** such that the wavefront of the liquid sample rises:

$$(46) \quad 0 \leq P_c = \frac{2}{W} \gamma_{la} \left(\frac{\cos \theta - \sin \alpha}{\cos \theta + \sin \alpha (\frac{\sin \alpha}{\sin \theta} - \cos \theta)} \right).$$

(47) A flow channel border is defined as an intersection of the opening **413**, the wall of the lower slot **411** and the surface of the connecting part **414c**. $\Delta P_{sub.c}$ is a liquid pressure at the wall of a flow channel border (i.e., the pressure of the liquid sample leaning against the lower slot **411** when located at the opening **413**), $\gamma_{sub.la}$ is a surface tension coefficient of the liquid sample, w is the width of the lower slot **411**, θ is the contact angle between the liquid sample and the wall of the lower slot **411**, α is a wavefront arc of the liquid sample and the flow channel border, and β is the contact angle between the opening **413** and the connecting part **414c**. Thus, $\Delta P_{sub.c}$ can be controlled by adjusting β . β is merely $0^\circ \leq \beta < 90^\circ$. When the angle β is 90° , $\Delta P_{sub.c}$ is smaller; however, a flow stop effect can be more easily achieved. In addition, when the relation $(\Delta P_{sub.L})$ of liquid pressure at the wall of the flow channel border and the capillary pressure of the liquid sample at the lower slot **411** satisfies $\Delta P_{sub.c} = \Delta P_{sub.L}$, the liquid sample in the lower slot **411** stops moving upward.

Fifth Embodiment

(48) FIG. **5** shows an exploded schematic diagram of the components of a biosensor strip **500** of the fifth embodiment. The biosensor strip **500** is similar to the biosensor strip **200** of the second embodiment, and differs in respect of a different exhaust layer. The exhaust layer **240** of the biosensor strip **200** has only one layer of air flow channel, whereas an exhaust layer has two layers of air flow channels. As shown, an exhaust layer **540** of the biosensor strip **500** includes a first exhaust set **510**, which has a first air flow channel **541** at a first layer **511**, a first upper cover **512** covering the first layer **511**, and a connecting hole **513** located at the first upper cover **512**; and a second exhaust set **520** disposed above the first exhaust set **510**, the second exhaust set **520** having a second air flow channel **542** at a second layer **521**, a second upper cover **522** covering the second layer, and an exhaust hole **523** located above the second upper cover **522**. The first air flow channel

541, the connecting hole **513**, the second air flow channel **542** and the exhaust hole **523** form an exhaust flow channel **550**. In this embodiment, the first air flow channel **541** and the second air flow channel **542** have the same flow channel pattern. In other embodiments, the first air flow channel **541** and the second air flow channel **542** have different flow channel patterns.

Sixth Embodiment

(49) FIG. **6A** shows an exploded schematic diagram of the components of a biosensor strip **600** of the sixth embodiment. FIG. **6B** shows a top-viewed perspective diagram (with the components overlaid) of an exhaust layer **640** of the biosensor strip **600** of the sixth embodiment. The biosensor strip **600** is similar to the biosensor strip **200** of the second embodiment, and differs in respect of a different exhaust layer. The exhaust layer **240** of the biosensor strip **200** has only one layer of air flow channel, whereas an exhaust layer of the fifth embodiment has three layers of air flow channels. Referring to both FIGS. **6A** and **6B**, the exhaust layer **640** includes a first layer **610**, a second layer **620** covering the first layer **610**, and a third layer **630** overing the second layer **620**. The exhaust layer **640** further includes a first air flow channel **641** distributed at the first layer **610**, a second air flow channel **642** distributed at the second layer **620**, and a third air flow channel **643** distributed at the third layer **630**. The first air flow channel **641**, the second air flow channel **642** and the third air flow channel **643** form the exhaust flow channel **650**. When a liquid sample enters the reaction flow channel **222**, the flow direction of the air in the exhaust flow channel **650** includes the following sequences: (1) flowing from the first layer **610**, passing through the second layer **620** and entering the third layer **630**; (2) then flowing from the third layer **630**, passing through the second layer **620** and returning to the first layer **610**; and (3) again flowing from first layer **610**, passing through the second layer **620** and entering the third layer **630**. The flow direction of the air in the exhaust flow channel **650** further includes repeating the sequences (2) and (3) and staying at the sequence (3). The biosensor strip **600** further includes an upper cover **660** covering the exhaust layer **640**. The upper cover **660** has an exhaust hole **661**, and the flow direction of the air in the exhaust flow channel **650** further includes, after the sequence (3), leading to the exhaust hole **661** of the upper cover **660** for discharging into outside. It can be understood with reference to FIG. **6A** that the first air flow channel **641** cannot directly communicate to the third air flow channel **643**; instead, the first air flow channel **641** communicates to the third air flow channel **643** via the second air flow channel **642**.

(50) Referring to both FIGS. **6A** and **6B**, the first layer **610**, the second layer **620** and the third layer **630** each respectively has a first air flow channel **641**, a second air flow channel **642** and a third air flow channel **643**, and each air flow channel includes a plurality of sections that are not in communication with one another. The exhaust flow channel **650** is configured so that the sequence (1), the sequence (2) and the sequence (3) each flows through different sections of the first air flow channel **641**. For example, when the sequence (1) flows through a section **641a** of the first air flow channel **641**, the subsequent sequence (2) flows through a section **641b** of the first air flow channel **641**, and the subsequent sequence (3) flows through a section **641c** of the first air flow channel **641**. Similarly, the exhaust flow channel **650** is configured so that the sequence (1), the sequence (2) and the sequence (3) each flows through different sections of the second air flow channel **642** from one another. For example, when the sequence (1) flows through a section **642a** of the second air flow channel **642**, the subsequent sequence (2) flows through a section **642b** of the second air flow channel **642**, and the subsequent sequence (3) flows through a section **642c** of the second air flow channel **642**. Similarly, the exhaust flow channel **650** is configured so that the sequence (1), the sequence (2) and the sequence (3) each flows through different sections of the third air flow channel **643** from one another. For example, when the sequence (1) flows through a section **643a** of the third air flow channel **643**, the subsequent sequence (2) flows through a section **643b** of the third air flow channel **643**, and the subsequent sequence (3) flows through a section **643c** of the third air flow channel **643**.

Seventh Embodiment

(51) FIG. 7 shows a schematic diagram of a biosensor strip **700** of the seventh embodiment, and shows only a perspective diagram of a reaction layer **720** and the substrate layer **210** overlaid with each other. The biosensor strip **700** is similar to the biosensor strip **200** of the second embodiment, and differ in that the biosensor strip **700** further includes, at the reaction layer **720**, a first auxiliary flow channel **710** and an auxiliary exhaust flow channel **730** capable of directly discharging into outside. The first auxiliary exhaust flow channel **710** is in communication with the reaction channel **221**. In other embodiment, the auxiliary exhaust flow channel **730** can be located at another layer. The first auxiliary exhaust flow channel **710** is to allow the flow of the liquid sample in a first partial region of the reaction flow channel **221** faster than another region in the presence of the exhaust flow channel **241**, wherein the first partial region includes an intersection of the reaction flow channel **221** and the first auxiliary exhaust flow channel **710**. The first partial region can further include a sampling port **222** extending from the reaction flow channel **221** to the intersection of the reaction flow channel **221** and the first auxiliary exhaust flow channel **710**, as a region **790** shown in FIG. 7. In this embodiment, the specific first partial region **710** in which the flow is to be accelerated is configured such that the liquid sample first comes into contact with the auxiliary exhaust hole **730** and then arrives at the through hole **250** corresponding to the exhaust flow channel **241**. Referring to FIG. 7, in this embodiment, the first auxiliary exhaust flow channel **710** allows the liquid sample to enter therein. When the liquid sample enters the sampling port **222**, the first auxiliary exhaust flow channel **710** determines the flow of the liquid sample, and when the first auxiliary exhaust flow channel **710** is with no air resistance function, the flow of the liquid sample is determined by the capillary action of the reaction flow channel **221**. Then, when the liquid sample fills the first auxiliary flow channel **710**, the function of the first auxiliary flow channel **710** becomes off (not in communication with the atmosphere). At this point, the flow of the liquid sample in the reaction flow channel **221** turns back to be controlled by the exhaust flow channel **241**. The first auxiliary exhaust flow channel **710** is a channel between the reaction flow channel **221** and the auxiliary exhaust hole **730** for directly discharging into outside. The flow of the liquid sample in the reaction flow channel **221** is more accelerated as the first auxiliary exhaust flow channel **710** gets shorter. It should be noted that, in this embodiment, the first auxiliary exhaust flow channel **710** is in communication with the atmosphere when the liquid sample has not yet entered the sampling port **222**, and such state thereof is referred to as an on state. In other embodiments, the first auxiliary exhaust flow channel **710** can be selectively disposed on any position of the reaction flow channel **221**, for example, at one or more of the plurality of reagent areas. In light of the above, it should be understood that the first auxiliary exhaust flow channel **710** can replace the exhaust flow channel **241** to determine the flow of the liquid sample in the first partial region of the reaction flow channel **221**.

(52) In another embodiment, apart from the first auxiliary exhaust flow channel **710**, the reaction layer **720** of the biosensor strip **700** further includes a second auxiliary exhaust flow channel (not shown). The second auxiliary exhaust flow channel is configured such that the liquid sample entering from the sampling port **22** first arrives at the first auxiliary exhaust flow channel **710**, then arrives at the second auxiliary exhaust flow channel by passing one section of the exhaust flow channel **241**, then further passing another section of the exhaust flow channel **241**, and finally arrives at the through hole **250** corresponding to the exhaust flow channel **241**. Different from the first auxiliary exhaust flow channel **710**, the second auxiliary exhaust flow channel is in a off state (not in communication with the atmosphere) when liquid sample has not entered the first auxiliary exhaust flow channel yet. After the liquid sample closes the first auxiliary exhaust flow channel **710**, the second auxiliary exhaust flow channel can be transformed from the off state to an on state (in communication with the atmosphere) at an appropriate timing, so as to accelerate the flow of the liquid sample in a second partial region of the reaction flow channel **241**. Various appropriate methods can be used to transform the second auxiliary exhaust flow channel from the off state to the on state. For example, upon the electrode unit **211** detecting that the liquid sample flows

through a specific position of the reaction flow channel **241**, the second auxiliary exhaust flow channel is activated to become open (on). In this embodiment, when the liquid sample has not entered the sampling port **222**, in the presence of the exhaust flow channel **241** (on), the first auxiliary exhaust flow channel **710** is on and the second auxiliary flow channel is off. When the liquid sample enters the sampling port **222** (entering the first partial region of the exhaust flow channel **241**), the first auxiliary exhaust flow channel **710** determines the flow of the liquid sample. When the liquid sample fills the first auxiliary exhaust flow channel **710**, the first auxiliary exhaust flow channel is off and the second auxiliary exhaust flow channel is not yet open (on), and the flow of the liquid sample at this point returns to the flow controlled by the exhaust flow channel **241**. Then, the second auxiliary exhaust flow channel turns to be on, and the liquid sample enters a second partial region of the exhaust flow channel **241**, and the second auxiliary exhaust flow channel at this point determines the flow of the liquid sample until the liquid sample fills the second auxiliary exhaust flow channel. When the liquid sample fills the first auxiliary exhaust flow channel **710** and also fills the second auxiliary exhaust flow channel, the flow of the liquid sample at this point returns to the flow controlled by the exhaust flow channel **241**. In light of the above, it should be understood that the second auxiliary exhaust flow channel can replace the exhaust flow channel **241** to determine the flow of the liquid sample in the second partial region of the reaction of the reaction flow channel **221**.

Eighth Embodiment

(53) FIG. **8** shows a schematic diagram of a biosensor strip **800** of the eighth embodiment, and depicts a top-viewed perspective diagram of only a reaction layer **820** and a substrate layer **810** overlaid with each other. The biosensor strip **800** differs from the biosensor strip **200** of the second embodiment in respect of the configuration of a reaction flow channel **821** of the reaction layer **820** and the configuration of the electrode unit of the substrate layer **810**. One feature of the eighth embodiment is that the path of the reaction flow channel **821** includes at least one bend, by which the flow of the liquid sample in the reaction flow channel **821** can be reduced. As shown, starting from a sampling port **822**, the path of the reaction flow channel includes a bend **821a**, a bend **821b**, a bend **821c**, a bend **821d**, a bend **821e**, a bend **821f** and a bend **821g**.

(54) Referring to FIG. **8**, another feature of the eighth embodiment is that the cross-section area of the reaction flow channel **821** changes with the different regions through which the reaction flow channel **821** flows. As shown, the cross-section area of the reaction flow channel **821** is not kept constant from the sampling port **822** to a terminal end **821t** of the reaction flow channel **821**, for example, regions B in the figure are all regions having greater cross-section areas. The flow of a specific first partial region of the reaction flow channel **821** can be fine-tuned by changing the cross-section area of the specific first partial region.

(55) FIG. **9A**, FIG. **9B** and FIG. **9C** show schematic diagrams of operating a conventional biosensor strip without an exhaust flow channel. The schematic diagram in FIG. **9A** shows that a liquid sample S has just been dripped into a reaction flow channel **920** of a biosensor strip **900**, and a timer T in FIG. **9A** indicates that the time is 04 seconds. The schematic diagram in FIG. **9B** shows that the liquid sample S has flown through more than $\frac{2}{3}$ of the reaction flow channel **920**, and a timer T in FIG. **9B** indicates that the time is 06 seconds. The schematic diagram in FIG. **9C** shows that the liquid sample S has arrived at the terminal end of the reaction flow channel **920**, and a timer T in FIG. **9C** indicates that the time is 07 seconds. It is demonstrated above that in the biosensor strip **900** without an exhaust flow channel, the time for the liquid sample S to fill the reaction flow channel is less than 10 seconds.

(56) FIG. **10A**, FIG. **10B**, FIG. **10C**, FIG. **10D**, FIG. **10E** and FIG. **10F** show schematic diagrams of operating a biosensor strip **1000** according to one embodiment of the present invention. The biosensor strip **1000** has an exhaust flow channel similar to that of the second embodiment. The shape of a reaction flow channel of the biosensor strip **1000** is the same as that of the biosensor strip **900**. The schematic diagram in FIG. **10A** shows that the liquid sample S has just been dripped

into a reaction flow channel **1020** of the biosensor strip **1000**, and a timer T in FIG. **10A** indicates that the time is 31 seconds. FIG. **10B**, FIG. **100**, FIG. **10D**, FIG. **10E** and FIG. **10F** sequentially show the process of the liquid sample S flowing through the reaction flow channel **1020** till the terminal end, wherein timers T in the respective schematic diagrams indicate that the time is 1 minute and 31 seconds, 2 minutes and 31 seconds, 3 minutes and 31 seconds, 4 minutes and 31 seconds, and 5 minutes and 44 seconds, respectively. It is demonstrated above that in the biosensor strip **1000** having an exhaust flow channel according to an embodiment of the present invention, the time for the liquid sample to fill the reaction flow channel **1020** is 5 minutes and 44 seconds. (57) FIG. **11A**, FIG. **11B** and FIG. **11C** show schematic diagrams of operating a biosensor strip **1100** according to an embodiment of the present invention. The biosensor strip **1100** has an exhaust flow channel similar to that of the second embodiment. The shape of a reaction flow channel of the biosensor strip **1100** is the same as that of the biosensor strip **1000**; however, the biosensor strip **1100** additionally includes an auxiliary exhaust flow channel **1120**. The schematic diagram in FIG. **11A** shows that the liquid sample S has just been dripped into a reaction flow channel **1020** of the biosensor strip **1100** and rapidly fills the auxiliary exhaust flow channel **1120**, and a timer T in FIG. **11A** indicates that the time is 06 seconds. FIG. **11B** and FIG. **11C** sequentially show the process of the liquid sample S flowing through the reaction flow channel **1020** till the terminal end, and timers T in the respective schematic diagrams indicate that the time is 1 minute and 31 seconds and 5 minutes and 31 seconds, respectively. It is demonstrated above that in the biosensor strip **1100** having an exhaust flow channel and an auxiliary exhaust flow channel according to an embodiment of the present invention, the time for the liquid sample to fill the reaction flow channel is 5 minutes and 31 seconds, which is less than the time needed by the embodiment in FIG. **10A** to FIG. **10F** (having an exhaust flow channel but not an auxiliary exhaust flow channel).

(58) Other specific forms can be used to embody the present invention without departing from the spirit or necessary features of the present invention, and the specific non-limiting embodiments in various aspects should be considered illustrative. Therefore, the scope of the present invention should be accorded with the appended claims but not as described in the foregoing description. All changes within equivalent meanings and ranges of the scope of the claims should be considered as within the scope of the claims.

SYMBOL DESCRIPTION

(59) f arrow/direction of air flow V1 volume V2 volume **100** biosensor strip **101** sampling end **110** substrate layer **120** reaction layer **121** reaction flow channel **130** separation layer **140** exhaust layer **141** exhaust flow channel **150** through hole **160** upper cover **161** exhaust hole **190** liquid sample **200** biosensor strip **201** sampling end **202** connecting end **210** substrate layer **211** electrode unit **220** reaction layer **221** reaction flow channel **221t** terminal end **222** sampling port **223a**, **223b**, **223c**, **223d** reagent area **224** reagent **230** separation layer **231** first notch **241** exhaust flow channel **242** second notch **250** through hole **260** upper cover **261** exhaust hole **262** third notch **300** biosensor strip **310** hydrophilic layer **311** first exposure opening **312** second exposure opening **313** third exposure opening **410** choke valve **411** lower slot **412** middle slot **413** open **413p** horizontal plane **414** bottom part **414c** connecting part **415** upper slot α wavefront arc β contact angle θ contact angle w width **500** biosensor strip **510** first exhaust set **511** first layer **512** first upper cover **513** connecting hole **520** second exhaust set **521** second layer **522** second upper cover **523** exhaust hole **540** exhaust layer **541** first air flow channel **542** second air flow channel **550** exhaust flow channel **600** biosensor strip **610** first layer **620** second layer **630** third layer **640** exhaust layer **641** first air flow channel **641a**, **641b**, **641c** sections **642** second air flow channel **642a**, **642b**, **642c** sections **643** third air flow channel **643a**, **643b**, **643c** sections **650** exhaust flow channel **660** upper cover **661** exhaust hole **700** biosensor strip **710** first auxiliary flow channel **720** reactive layer **730** auxiliary exhaust flow channel **790** region **800** biosensor strip **810** substrate layer **811** substrate layer **820** reactive layer **821** reaction flow channel **821a**, **821b**, **821c**, **821d**, **821e**, **821f**, **821g** bend B region **821t** terminal end **821f** bend **822** sampling port **900** biosensor strip **920** reaction flow channel S

liquid sample T timer **1000** biosensor strip **1020** reaction flow channel **1100** biosensor strip **1120**
auxiliary exhaust flow channel

Claims

1. A biosensor strip, comprising: a reaction layer, comprising a reaction flow channel; a separation layer, located above the reaction layer and covering the reaction layer; an exhaust layer, located above the separation layer, the exhaust layer comprising an exhaust flow channel and an exhaust hole configured to discharge air into outside and be in communication with the exhaust flow channel; and a through hole, passing through the separation layer to communicate with the exhaust flow channel and the reaction flow channel; wherein, the reaction flow channel comprises a sampling port, the reaction flow has a capillary structure configured to allow a liquid sample to enter the reaction flow channel through the sampling port, and the exhaust flow channel together with the exhaust hole are configured to reduce a flow of the liquid sample in the reaction flow channel provided by a capillary action; and wherein, an air flow in the exhaust flow channel is configured to be smaller than the flow of the liquid sample provided by the capillary action.
2. The biosensor strip according to claim 1, wherein a resistance of air in the exhaust flow channel is $R_{\text{sub.G}}$, a resistance difference between the liquid sample and air in the reaction flow channel is $\Delta R_{\text{sub.L}}$, and $R_{\text{sub.G}} \gg \Delta R_{\text{sub.L}}$.
3. The biosensor strip according to claim 1, wherein the exhaust flow channel is distributed in a wave manner on the exhaust layer.
4. The biosensor strip according to claim 1, wherein the exhaust flow channel has a lateral cross section perpendicular to an extension direction of the exhaust flow channel, the lateral cross section has a long diameter and a short diameter, and a ratio of the long diameter to the short diameter is greater than 5.
5. The biosensor strip according to claim 1, wherein the through hole comprises a choke valve, the choke valve having a lower slot in communication with the reaction flow channel; and a middle slot located above the lower slot, wherein the lower slot has an opening in communication with the middle slot, the opening defining a horizontal plane, and the middle slot has a bottom part, the bottom part being formed with a connecting part to connect the opening, wherein a surface of the connecting part is not higher than the horizontal plane.
6. The biosensor strip according to claim 4, wherein the through hole comprises a hydrophobic inner wall.
7. The biosensor strip according to claim 1, wherein the exhaust layer comprises: a first exhaust set, comprising a first air flow channel at a first layer, a first upper cover covering the first layer, and a connecting hole located at the first upper cover; and a second exhaust set disposed above the first exhaust set, the second exhaust set comprising a second air flow channel at a second layer, a second upper cover covering the second layer, and an exhaust hole located at the second upper cover; wherein, the first air flow channel, the connecting hole, the second air flow channel and the exhaust hole form the exhaust flow channel.
8. The biosensor strip according to claim 1, wherein the exhaust layer comprises a first layer, a second layer covering the first layer, and a third layer covering the second layer; when a liquid sample enters the reaction flow channel, a flow direction of air in the exhaust flow channel comprises following sequences: (1) flowing from the first layer, passing through the second layer, and entering the third layer; (2) then flowing from the third layer, passing through the second layer, and returning to the first layer; and (3) again flowing from the first layer, passing through the second layer and entering the third layer.
9. The biosensor strip according to claim 1, the reaction layer comprising at least one auxiliary exhaust flow channel in communication with the reaction flow channel, wherein the auxiliary exhaust flow channel replace the exhaust flow channel to determine a flow of the liquid sample in a

partial region of the reaction flow channel, and wherein the partial region further comprises an intersection of the reaction flow channel and the auxiliary exhaust flow channel.

10. The biosensor strip according to claim 9, wherein the partial region comprises a range from the sampling port of the reaction flow channel extended to the intersection of the reaction flow channel and the auxiliary exhaust flow channel.
